Pro Football Sports Science Research Update

April 2025

A multilevel model with heterogeneous variances for snap timing in the National Football League. arXiv.2502.16313, 2025.

Associations between match-play characteristics and environmental temperatures in 4 professional football leagues. European Journal of Sport Science, 25(3): e12256, 2025.

Comparing time to recovery between initial and repeat concussion in athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Data analytics practices and reporting strategies in senior football: Insights into athlete health and performance from over 200 practitioners worldwide. Science and Medicine in Football, Ahead of Print: 1-17, 2025.

Early warning system for player recovery? A series of case studies illustrating the application of individualized adaptive reference ranges in the longitudinal blood monitoring of English Premier League soccer players. Biomarkers, Ahead of Print: 1-28, 2025.

Effects of resistance and speed on electromyographic activity of thigh and gluteal muscles in elite athletes throughout resisted sprint running. Medicine and Science in Sports and Exercise, Ahead of Print: 1-53, 2025.

Evolving the control-chaos continuum: Part 1 – Translating knowledge to enhance on-pitch rehabilitation. Journal of Orthopaedic & Sports Physical Therapy, 55(2): 78-88, 2025.

Evolving the control-chaos continuum: Part 2 – Shifting "attention" to progress on-pitch rehabilitation. Journal of Orthopaedic & Sports Physical Therapy, 55(3): 162-172, 2025.

Hamstrings muscle architecture and morphology following 6 weeks of an eccentrically-biased Romanian deadlift or Nordic hamstring exercise intervention. Medicine and Science in Sports and Exercise, Ahead of Print: 1-51, 2025.

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Health-promoting behaviors and their associations with factors related to well-being among former National Football League players: An NFL-LONG Study. Journal of Athletic Training, 60(2): 185-197, 2025.

Higher level of communication between the medical staff and the performance staff is associated with a lower hamstring injury burden: A substudy on 14 teams from the UEFA Elite Club Injury Study, BMJ Open Sport & Exercise Medicine, 11(1): e002182, 2025.

Immediate effects of overnight long-haul travel on physical and cognitive performance and sleep in professional male Rugby Union players (A 2-part study). International Journal of Sports Physiology and Performance, Ahead of Print: 1-10, 2025.

Influence of running speed and time post-operatively on lower extremity work in collegiate athletes after anterior cruciate ligament reconstruction. Medicine and Science in Sports and Exercise, Ahead of Print: 1-40, 2025.

Injection of amniotic membrane and umbilical cord particulate for muscle and ligament tears in collegiate football athletes: A single-center, retrospective study. Orthopaedic Research and Reviews, 16: 295-301, 2024.

Key performance indicators for college American football starters: An exploratory study. Journal of Functional Morphology and Kinesiology, 10: 19, 2025.

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Toxic leadership in high-performance sports and its consequences for mental health and performance: A scoping review. International Review of Sport and Exercise Psychology, Ahead of Print: 1-28, 2025.

Weighted approaches to assessing injury burden in team sports. Journal of Sports Analytics, 10(4): 1-9, 2025.

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A multilevel model with heterogeneous variances for snap timing in the National Football League. arXiv.2502.16313, 2025.

Abstract - Player tracking data have provided great opportunities to generate novel insights into understudied areas of American football, such as pre-snap motion. Using a Bayesian multilevel model with heterogeneous variances, we provide an assessment of NFL quarterbacks and their ability to synchronize the timing of the ball snap with pre-snap movement from their teammates. We focus on passing plays with receivers in motion at the snap and running a route, and define the snap timing as the time between the moment a receiver begins motioning and the ball snap event. We assume a Gamma distribution for the play-level snap timing and model the mean parameter with player and team random effects, along with relevant fixed effects such as the motion type identified via a Gaussian mixture model. Most importantly, we model the shape parameter with quarterback random effects, which enables us to estimate the differences in snap timing variability among NFL quarterbacks. We demonstrate that higher variability in snap timing is beneficial for the passing game, as it relates to facing less havoc created by the opposing defense. We also obtain a quarterback leaderboard based on our snap timing variability measure, and Patrick Mahomes stands out as the top player.

Associations between match-play characteristics and environmental temperatures in 4 professional football leagues. European Journal of Sport Science, 25(3): e12256, 2025.

Abstract - This study investigated the association between environmental temperature and match-play characteristics (shooting, passing, dribbling and defending) in four professional football leagues. Twenty-seven performance indicators (PI's) were collated from 1585 matches from the German Bundesliga 1 and 2, Spanish La Liga and Australian A-League. Environmental data were obtained for dry-bulb temperature (T) and wet-bulb globe temperature (WBGT) retrospectively from public sources. For each league, linear regressions were used to determine relationships between PI's and T and WBGT and linear mixed models were used to determine those associations across all four leagues. Individual leagues showed varying associations between a collection of PI's and environmental measures. When combining the four leagues' match data, 8 of the 17 investigated parameters were associated with T and WBGT (p <

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0.002). Passes, especially short passes, were reduced in higher T (-2.3 [-3.1 to -1.5] and $p < 0.001$) and WBGT (-3.1 [-4.0 to -2.1] and $p < 0.001$), alongside an increase in the success rate of passes (0.06 [0.02-0.09] and $p \leq 0.001$). The number of passes into the opponent's final third was reduced for both T (-0.18 [-0.25 to -0.05] and $p = 0.001$) and WBGT (-0.17 [-0.28 to-0.05] and $p = 0.002$), but the number of key passes leading to a shot or goal was not associated with T or WBGT ($p \geq 0.67$). The number of touches, take-ons and turnovers were reduced in higher T and WBGT (all $p < 0.001$). Accordingly, in higher heat stress, match actions, especially those performed at high volumes, are reduced. Therefore, teams should expect a possibly altered match play and may consider adapting tactical or heat-mitigating strategies to counter these effects.

Comparing time to recovery between initial and repeat concussion in athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Objective: Compare time to recovery between initial and repeat concussions.

Design: Retrospective review of electronic medical record.

Setting: An interdisciplinary concussion clinic.

Patients: Fifty patients (14.4 years old [±2.5]; 24 [48%] female) who sustained >1 concussion between 2016 and 2020 and were evaluated within 30 days of their concussion.

Intervention: Injuries were divided into first, second, and/or last concussions.

Main outcome measures: The primary outcome was time to recovery, as defined by clearance for return to sport. This was compared between first versus second concussion (primary analysis) and first versus last concussion (secondary analysis) using Cox regression. Multivariable models accounted for time from concussion to initial clinic visit and Sport Concussion Assessment Tool symptom severity score at initial encounter. Sensitivity analyses explored potential influences of initial encounter timing and interval between concussions.

Results: Time to recovery did not differ between initial and repeat concussion in univariate or multivariable models (HR = 1.02, 95% CI, 0.65-1.62, P = 0.92; HR = 1.01, 95% CI, 0.64-1.62, P = 0.91 for

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multivariable models comparing first (ref) versus second and first (ref) versus last concussion, respectively). Similar results were obtained in sensitivity analyses stratifying by time to initial visit (≤7 days vs >7 days) or time between injuries (≤1 year vs >1 year).

Conclusions: There were no differences in time to recovery between initial and repeat concussions sustained by the same athlete. A nonsignificant trend of shorter recovery after repeat concussions was observed for patients presenting within 1 week of injury and when repeat injuries occurred within 1 year.

Data analytics practices and reporting strategies in senior football: Insights into athlete health and performance from over 200 practitioners worldwide. Science and Medicine in Football, Ahead of Print: 1-17, 2025.

Abstract - Despite the rise of data generation in football, the expertise of data analytics within the sport is relatively underdeveloped. To further understand the landscape, a cross-sectional, observational study design was used to survey practitioners in senior, professional, or semi-professional football. Areas of interest included the personnel involved (the 'who'), the data collected (the 'what'), and the analytical techniques employed (the 'how'). A total of 206 practitioners completed an online survey, with representation from all six FIFA confederations. Of the 206 respondents, 86% were male, 13% female, and 1% preferred not to disclose their gender. Respondents were categorized as working in either the performance (73%), data (18%), or medical (9%) department. Heterogeneity was observed in responses across all departments regarding training load metrics, outcome metrics, methodological attributes, and measurement properties. Evidence sources used prior to implementing a new metric varied between departments, with performance (63%) and medical (67%) staff relying on professional industry and/or community, while data staff (57%) utilized more in-house projects. The analytical approach used most frequently was exploratory data analysis (90%), with modelling, forecasting, and predicting the least frequent (54%). Respondents reported using a mix of solutions for data storage, aggregating and analyzing, and reporting and visualizing data. Spreadsheets were cited as a popular solution for data wrangling and reporting tasks. The findings provide an overview of current data ecosystems and information systems in modern football organizations. These results can be used to

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improve data analytics service provision in football by helping identify areas for development and progression.

Early warning system for player recovery? A series of case studies illustrating the application of individualized adaptive reference ranges in the longitudinal blood monitoring of English Premier League soccer players. Biomarkers, Ahead of Print: 1-28, 2025.

Abstract - Blood biomarkers can provide objective insight into a player's physiological state of recovery. In the context of elite football, the utilization of a point-of-care (POC) test for measuring inflammation via capillary sampling, facilitates frequent data acquisition in order to assess recovery status within game-to-game micro-cycles. The smaller within-subject variability compared to the variability between subjects limits the practical use of population-based reference ranges. Individualized approaches to biomarker monitoring may therefore be of higher potential value in assessing player health and recovery. We aimed to explore the application of individualized adaptive reference ranges (IARR) in English Premier League (EPL) soccer players using a POC biomarker for C-Reactive Protein (CRP) as a marker of inflammation. Using historical data collected from players' CRP values during the 2019-2020 season, we evaluated the effectiveness of static and IARR in identifying abnormal values and reported sensitivity and specificity at a 5% significance level. Our analysis confirmed that individualized monitoring with personalized adaptive reference ranges is more effective in identifying true abnormalities compared to PBRR (87.8% vs 86.2% for pre-match CRPs and 85.2% vs 81.5% for post-match CRPs), particularly when the intra-individual variability is considerably lower than inter-individual variability. The application of IARR for blood monitoring data could assist the practitioner in identifying periods where a player may require performance (e.g. workload management and recovery practices) or medical support from the multi-disciplinary team. However, IARR serve more as an early warning system than a diagnostic tool. Thus, significant care is needed to prevent misuse and misinterpretation when implementing this strategy in real-world settings.

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Effects of resistance and speed on electromyographic activity of thigh and gluteal muscles in elite athletes throughout resisted sprint running. Medicine and Science in Sports and Exercise, Ahead of Print: 1-53, 2025.

Purpose: The application of horizontal resistance is key in sprint training to modulate velocity and promote adaptations. However, the interaction between velocity and resistance on muscle excitation remains insufficiently studied. This study assessed the electromyographic (EMG) activity of thigh and gluteal muscles in response to varying velocity and resistance during sprinting.

Methods: Thirty-seven elite athletes (27 females and 10 males) performed two trials under three conditions: a 40-meter maximal sprint without resistance and two resisted sprints applied by a robotic device with resistive forces equivalent to 25% and 75% of body mass in a sled condition. EMG activity was recorded from eight muscles in the quadriceps, hamstrings, and gluteal muscle groups for both lower limbs. Generalized mixed models were used to analyze average EMG changes with increasing velocity as a function of resistance. Statistical parametric mapping was used to assess the changes within the stance and swing phases as a function of sprint phase and resistance.

Results: Hamstring EMG activity increased with increasing velocity (+17.9% in stance phase without resistance) and decreased as resistance increased, with opposite effects observed in quadriceps muscles. Higher gluteal EMG activity was found during the late swing phase with increasing velocity and decreasing resistance. The biceps femoris long head exhibited the largest EMG increase with velocity (+23% in the stance phase) among hamstrings, while the semitendinosus showed the highest increase with reduced resistance (+27% in the stance phase).

Conclusions: External resistance and running speed influence thigh muscle activity differently, even within the same muscle group, likely due to distinct muscle architecture and function. These insights should be considered when designing sprint training programs to target specific muscle groups, avoid muscle overloading, and account for inter-individual differences to optimize performance and reduce injury risk.

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Evolving the control-chaos continuum: Part 1 – Translating knowledge to enhance on-pitch rehabilitation. Journal of Orthopaedic & Sports Physical Therapy, 55(2): 78-88, 2025.

Background: On-pitch rehabilitation is a crucial part of returning to sport after injury in elite soccer. The control-chaos continuum (CCC) initially offered a framework for practitioners to plan on-pitch rehabilitation, focusing on physical preparation and sport specificity. However, our experiences with the CCC, combined with recent research in injury neurophysiology, point to a need for an updated model that integrates practice design and physical-cognitive interactions.

Clinical Question: What are the insights from injury neurophysiology, soccer performance, and coaching science needed to update the CCC and improve the planning, delivery, and progression of on-pitch rehabilitation in elite soccer? KEY RESULTS: Drawing on extensive experience in elite sport, we explain how recent research on neurophysiological recovery from injury, game models, and practice design has been applied to update the CCC and evolve the existing framework.

Clinical Application: The evolution of the CCC expands on the original model to enhance planning, delivery, and progression of on-pitch rehabilitation. The updated framework incorporates elements of visual cognition, attentional challenges, decision-making, and progressive representation of the game model to enhance sport-specific preparation for returning to sport.

Evolving the control-chaos continuum: Part 2 – Shifting "attention" to progress on-pitch rehabilitation. Journal of Orthopaedic & Sports Physical Therapy, 55(3): 162-172, 2025.

Background: On-pitch rehabilitation is key to supporting return to sport in elite soccer (football). The control-chaos continuum (CCC) guides practitioners through the sport-specific components of return to sport. There is a need to update the framework with recent research in injury neurophysiology, soccer performance, and coaching science.

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Clinical Question: How do practitioners integrate the insights from injury neurophysiology, soccer performance, and coaching science discussed in part 1 of this 2-part series into an updated version of the CCC, to enhance the planning and execution of on-pitch rehabilitation for elite soccer players?

Key Results: The revised CCC framework emphasizes the design and delivery of progressive training in increasingly chaotic conditions. The updated framework supports practitioners to incorporate elements of visual-cognitive challenges, attentional challenges, decision making, and progression representation of the game model when players are preparing to return to sport.

Clinical Application: The updated CCC outlines training progression from High Control, which involves returning to on-pitch linear running, to High Chaos, which simulates the team environment at game speed. The High Chaos phase can be tailored to support players during their reintegration into team training. The model is adaptable for both short- and long-term injuries, integrating physical-cognitive load monitoring and strength and power diagnostics to enhance decision making throughout return to sport.

Hamstrings muscle architecture and morphology following 6 weeks of an eccentrically-biased Romanian deadlift or Nordic hamstring exercise intervention. Medicine and Science in Sports and Exercise, Ahead of Print: 1-51, 2025.

Purpose: This study investigated the effects of a 6-week eccentrically-biased training intervention between Romanian deadlift (RDL) and Nordic hamstring exercises (NHE) on regional hamstring muscle architecture and morphology.

Methods: Participants were randomly allocated to an RDL or NHE intervention group and trained twice per week for 6 weeks followed by a 2-week detraining period. Biceps femoris long head (BFlh) architecture (fascicle length [FL] and pennation angle [PA]) and muscle thickness (MT) were measured using extended-field-of-view ultrasound. Anatomical cross-sectional areas (ACSA) of BFlh, semimembranosus, and semitendinosus were also assessed via ultrasound. Muscle architecture and

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morphology were measured proximally, in the middle, and distally along the thigh. Separate linear mixed effects models were run for each outcome.

Results: Thirty-two healthy participants (15 females, 17 males) aged 20.8 (0.9) years were enrolled and allocated to RDL (n = 16) or NHE (n = 16) groups. Following the intervention, BFlh FL increased (0.80 [0.6, 1.0] cm (9%), $p < 0.001$), while PA decreased uniformly across all regions (-1.2° [-1.5, -0.9] (10%), $p < 0.001$) and MT did not change ($p = 0.29$). Both BFlh FL and PA returned to baseline after detraining. No differences were observed between groups for changes in FL ($p = 0.55$), PA ($p = 0.74$), or MT ($p = 0.48$). Hamstrings ACSA increased after the intervention (0.78 cm2 [0.45, 1.11] (10%), $p < 0.001$) and remained elevated after detraining. There were no significant differences in ACSA between groups after the intervention ($p = 0.60$).

Conclusions: Changes in FL, PA, and ACSA are consistent between an eccentrically-biased RDL and NHE intervention. These findings provide practitioners choice in exercise selection for eliciting adaptations relevant to hamstring injury prevention.

Health-promoting behaviors and their associations with factors related to well-being among former National Football League players: An NFL-LONG Study. Journal of Athletic Training, 60(2): 185-197, 2025.

Context: Understanding former professional football players' engagement with health-promoting behaviors (physical exercise, high-quality diet, and good sleep hygiene) will be helpful for developing lifestyle interventions to improve their feelings of well-being, a relatively understudied facet of health among this population.

Objective: Examine associations among health-promoting behaviors and subjective outcomes related to well-being among former National Football League (NFL) players.

Design: Cross-sectional.

Setting: Online or hard-copy survey.

Patients or other participants: Former NFL players.

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Main outcome measure(s): Self-reported health-promoting behaviors (exercise frequency, diet quality, and sleep duration and disturbance) and factors related to well-being (Patient-Reported Outcomes Measurement Information System - Meaning and Purpose [MP], Self-Efficacy, and Ability to Participate in Social Roles and Activities [SRA]). Multivariable linear regression models were fit for each well-being-related factor with health-promoting behaviors as explanatory variables alongside select demographic, behavioral, and functional covariates. Models were fit for the full sample and separately for individual age groups: <30 years, 30 to 39 years, 40 to 49 years, 50 to 59 years, 60 to 69 years, and 70+ years.

Results: A total of 1784 former NFL players (aged 52.3 ± 16.3 years) completed the survey. Lower sleep disturbance was associated with better MP (β [standard error] = -0.196 [0.024]), Self-Efficacy (β [standard error] = -0.185 [0.024]), and SRA (β [standard error]= -0.137 [0.017]) in the full sample and almost all the individual age groups. More frequent moderate-to-vigorous exercise was associated with higher MP (β [standard error] = 0.068 [0.025]) and SRA (β [standard error] = 0.151 [0.065]) in the full sample and with better MP, Self-Efficacy, and SRA among select middle-aged groups (between 40 and 69 years old). Diet quality, resistance training exercise frequency, other wellness activity frequency, and sleep duration were not associated with well-being-related factors in the full group, and sparse significant associations were observed in individual age group models.

Conclusion: Lower sleep disturbance and more frequent moderate-to-vigorous exercise frequency may be important targets for improving overall health and well-being among former NFL players.

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Higher level of communication between the medical staff and the performance staff is associated with a lower hamstring injury burden: A substudy on 14 teams from the UEFA Elite Club Injury Study, BMJ Open Sport & Exercise Medicine, 11(1): e002182, 2025.

Objectives: External risk factors connected to club, team and coaching are believed to be important in the causation of hamstring injuries, but little is known about the preventive measures used. The objective was to analyze the association between preventive factors and hamstring muscle injury burden.

Methods: 14 teams participated in the Union of European Football Associations Champions/Europa Leagues from 2019/2020 to 2022/2023. An open question was sent out in January 2023 to the four teams with the lowest hamstring muscle injury burden, asking why their teams had successfully avoided hamstring muscle injuries and which preventive methods they had used. A questionnaire that investigated the use of these methods was sent to all 14 teams in February 2023. A multiple linear regression model was applied, using injury burden as an outcome variable and the questionnaire responses as possible explanatory variables possibly adjusted for team and season.

Results: The preventive methods included factors such as the level of communication between the coach, the medical staff and the performance staff. The adjusted multiple linear regression model indicated that communication between the medical staff and the performance staff was negatively associated with an increased hamstring muscle injury burden, where for every 'higher' grade of the communication variable, the average injury burden decreased by 4.1 (95% CI -8.2 to -0.04, p=0.048) days per 1000 hours.

Conclusions: Better communication between the medical and the performance staff and continuous involvement of the medical staff in load management was associated with a lower hamstring injury burden in male professional football teams.

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Immediate effects of overnight long-haul travel on physical and cognitive performance and sleep in professional male Rugby Union players (A 2-part study). International Journal of Sports Physiology and Performance, Ahead of Print: 1-10, 2025.

Background: Travel can disrupt sleep and reduce perceived recovery in athletes. While the impact of travel on physical performance has been studied, its effect on cognitive performance remains unexplored.

Purpose: This study investigated the consequences of overnight long-haul travel on physical (part I) and cognitive performance (part II) in professional rugby players traveling overseas for competition.

Methods: In part I (n = 24), vertical-jump performance was assessed across 5 mornings surrounding travel (HOME, TRAVEL, and AWAY 1, 2, and 3). In part II (n = 24), mental fatigue and psychomotor vigilance (defined as the ability to respond to stimuli effectively) were assessed across 3 days (HOME, TRAVEL, and AWAY 1). Sleep was monitored on all nights preceding performance assessment using activity monitoring.

Results: Compared with HOME, TRAVEL reduced sleep quantity (part I. −219 [112] min; part II, −334 [233] min; both $P < .001$) and elevated in-flight mental fatigue (part II, +12 [18]; $P < .05$). Psychomotor vigilance was impaired during TRAVEL (+8% [6%] in valid response latencies ≤500 milliseconds; $P < .001$), and reactive jump strength was reduced at AWAY 1 (−7% [10%] in drop jump; $P < .001$).While psychomotor vigilance was restored by AWAY 1 (after hotel sleep), reactive jump strength remained diminished until AWAY 3 (−12% [9%] in drop jump compared with HOME; $P < .001$) after travel.

Conclusion: For teams traveling on long-haul journeys overnight for competition, it is recommended to arrive more than 3 days before matches to enable sufficient physical and cognitive recovery at the new destination to optimize performance away from home.

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Influence of running speed and time post-operatively on lower extremity work in collegiate athletes after anterior cruciate ligament reconstruction. Medicine and Science in Sports and Exercise, Ahead of Print: 1-40, 2025.

Purpose: Following anterior cruciate ligament reconstruction (ACLR), individuals present with significant and persistent deficits in surgical knee biomechanics during running. Little is known regarding the role of the hip and ankle in compensating for these knee-specific deficits. The purpose of this study was to characterize how limb (surgical and non-surgical), time post-operatively (3.5-7 months [EARLY], 8-13 months [LATE]), and running speed influence lower extremity joint-specific and total work during running post-ACLR.

Methods: Fifty-six Division I collegiate athletes post-ACLR (EARLY: n = 41, LATE: n = 42, Both: n = 27) completed running analyses at 2.68, 2.95, 3.35, 3.80 and 4.47 m/s or up to their maximum comfortable speed. Linear mixed effects models assessed the influence of limb, speed, time post-ACLR, and their interactions on hip, knee, ankle, and total negative and positive work.

Results: Surgical limb hip positive work was greater (0.044 J/kg [0.015, 0.074], $p < 0.001$) than the non-surgical limb EARLY. Surgical limb ankle negative (-0.054 J/kg [-0.093, -0.015], $p = 0.003$) and positive work (-0.115 J/kg [-0.168, -0.063], $p < 0.001$) were less than the non-surgical limb EARLY. Surgical limb knee negative (EARLY: -0.339 J/kg [-0.382, -0.296], $p < 0.001$; LATE: -0.222 J/kg [-0.265, -0.180], $p < 0.001$) and positive work (EARLY: -0.214 J/kg [-0.246, -0.182], $p < 0.001$; LATE: -0.142 J/kg [-0.174, -0.110], $p < 0.001$) were less than the non-surgical limb. Total negative and positive work followed the same pattern as the knee.

Conclusions: Athletes post-ACLR exhibited greater hip positive work and lesser ankle positive and negative work during running of the surgical limb EARLY, that resolved LATE. Both knee and total negative and positive work of the surgical limb were substantially less than the non-surgical limb, which improved from EARLY to LATE, but between-limb asymmetries remained.

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Injection of amniotic membrane and umbilical cord particulate for muscle and ligament tears in collegiate football athletes: A single-center, retrospective study. Orthopaedic Research and Reviews, 16: 295-301, 2024.

Background: Football is a leading cause of sports-related injuries, accounting for more than half of time-loss injuries in men's collegiate sports. Amniotic membrane and umbilical cord (AM/UC) powder injection is a novel treatment that may aid in functional recovery and return to play in collegiate athletes due to its anti-inflammatory, anti-scarring, and pro-regenerative properties.

Methods: A single-center, retrospective chart review was performed on consecutive college football players who sustained acute muscle or ligament tear and were subsequently treated with 50 or 100 mg injection of AM/UC powder (Clarix Flo, BioTissue, Miami, FL). Data collection included patient demographics, diagnosis, grade and extent of injury, position of the player, time to return to play (in days), and complications.

Results: Ten athletes with tears of the medial collateral ligament (n=3), hamstring (n=6), and abdomen (n=1) were included for analysis. The majority (n=6) of tears were partial (grade 2), with partial to full tears (grade 2±3) noted in 2 patients, and a complete tear (grade 3) noted in 2 patients. Athletes returned to play 29.6 ± 15.2 days post-injury, with 80% of athletes returning to play within 4 weeks. No complications or re-injuries occurred during the follow-up period of 6 months.

Conclusion: Injection of AM/UC powder is a safe treatment that enables quick return to play in this cohort of football players suffering from muscle or tendon tears. Prospective, randomized studies are warranted to verify whether injection of AM/UC can hasten return to play compared to the current standard treatment.

Level of evidence: IV.

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Key performance indicators for college American football starters: An exploratory study. Journal of Functional Morphology and Kinesiology, 10: 19, 2025.

Objectives: The purpose of this study was twofold: (a) to profile body composition and physical fitness characteristics of collegiate American football starters and (b) to examine differences in key performance indicators across position groups. These indicators included select measures of body composition, joint kinematics, as well as muscular strength and power.

Methods: Sixteen National Collegiate Athletic Association (NCAA) Division-II American football athletes (age: 22.25 ± 1.1 years; height: 183.75 ± 7.8 cm; and body mass: 97.22 ± 20.39 kg) volunteered to participate in this study. A Kruskal– Wallis one-way analysis of variance by ranks test with Dunn test post-hoc adjustments was used to examine position differences between Line (n = 3), Big Skill (n = 6), and Skill (n = 7) position groups with α priori set at $p < 0.05$.

Results: The findings of this study suggest that significant differences in body composition ($p = 0.004$), muscular strength ($p = 0.01$), and muscular power ($p = 0.03$) exist between position groups. However, no significant differences were observed in joint kinematics as assessed by the bilateral squat test ($p > 0.05$).

Conclusions: Therefore, key findings from this study suggest that although significant differences in body composition, muscular strength, and muscular power exist, an emphasis should be placed on the regular assessment, development, and maintenance of optimal joint kinematics within collegiate American football populations as this appears to be a shared key performance indicator among starters.

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Novel noninvasive imaging techniques to assess structural, functional, and material properties of tendon, ligament, and cartilage: A narrative review of current concepts. The Orthopaedic Journal of Sports Medicine, 13(2): 1-13, 2025.

Background: Novel noninvasive imaging modalities such as quantitative magnetic resonance imaging (qMRI) and shear wave elastography (SWE) allow for assessment of soft tissue microstructure and composition, which ultimately may be associated with functional and material properties.

Purpose: To provide a narrative review of the scientific techniques and clinical applications of qMRI and SWE for the evaluation of soft tissue about the knee and shoulder, including the meniscus, the anterior cruciate ligament (ACL), and the rotator cuff.

Study design: Review.

Methods: A literature search was performed in October 2022 via PubMed using the following keywords: "quantitative MRI tendon," quantitative MRI ligament,""quantitative MRI cartilage," or "shear wave elastography tendon." Only articles related to clinical applications were included in this review.

Results: Conventional imaging techniques, including standard morphologic magnetic resonance imaging (MRI) and ultrasound imaging, have limited ability to evaluate the material and functional properties of soft tissue; qMRI builds on the limitations of conventional morphologic MRI by allowing for detection of early articular cartilage changes, differentiation of healed versus unhealed meniscal tissue, and quantification of ACL graft maturity. SWE can evaluate the material properties of rotator cuff and Achilles tendons after injury, which may provide insight into both the chronicity and the healing status of the aforementioned injuries.

Conclusion: Our review of the literature showed that quantitative imaging techniques, including qMRI and SWE, may both improve early detection of pathology and aid in comprehensive evaluation after treatment.

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Omega-3 fatty acids and muscle strength – Current state of knowledge and future perspectives. Nutrients, 16: 4075, 2024.

Abstract - Omega-3 polyunsaturated fatty acids (n-3 PUFAs), specifically the most biologically active (eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA)), have well-documented health promoting effects, including, but not limited to, controlling inflammatory processes in the human body and supporting cardiovascular and cognitive health and visual processes. There is also some evidence pointing to the potential of EPA and DHA to preserve and/or enhance muscle strength. Muscle strength plays a crucial role in preventing age-related decline in skeletal muscle mass and function and the subsequent impaired functionality in the daily lives of the elderly. It also has a significant role in physical performance by aiding in the effectiveness of training elements, such as vertical jumps, sprinting, changes of direction, rate of force development, and anaerobic conditioning. Previous studies have indicated that supplementation with n-3 PUFAs may have a small but significant effect on preserving and/or increasing muscle strength in both healthy adults and in elderly. However, the number of studies published on this topic is limited. The goal of this narrative review is to summarize the effects of EPA and DHA supplementation on muscle strength and identify the limitations of previous studies that, if addressed, will help confirm or exclude the beneficial effects of n-3 PUFAs on muscle strength in humans.

Physiological Resilience: What is it and how might it be trained? Scandinavian Journal of Medicine & Science in Sports, 35(3): e70032, 2025.

Abstract - Physiological resilience has recently been recognized as an additional factor that influences endurance exercise performance. It has thus been incorporated into a modified, contemporary version of "the Joyner model" which acknowledges that start-line values of $\dot{V}O_2$max, efficiency or economy, and metabolic thresholds are prone to deterioration, often with appreciable interindividual variability, during prolonged endurance exercise. The physiological underpinnings of resilience are elusive and sports physiologists are presently concerned with developing practical testing protocols which reflect an athlete's resilience characteristics. It is also important to consider why some athletes are more resilient than others and whether resilience can be enhanced-and, if so, which training programs or specific

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training sessions might stimulate its development. While data are scant, the available evidence suggests that training consistency and the accumulation of relatively large volumes of training over the longer-term (i.e., several years) might promote resilience. The inclusion of regular prolonged exercise sessions within a training program, especially when these include bouts of high-intensity exercise at race pace or above or a progressive increase in intensity in the face of developing fatigue, might also represent an effective means of enhancing resilience. Finally, resistance training, especially heavy strength and plyometric training, appears to have positive effects on resilience. Considerations of training for resilience, alongside other more established physiological determinants of performance, will likely be important in the long-term development of successful endurance athletes.

Prevention of concussion and long-term effects of repetitive traumatic brain injury: Unanimous consensus from the Cantu Concussion Summit. PM & R, Ahead of Print: 1-4, 2025.

Objective: The Cantu Concussion Summit aimed to gather clinicians and researchers to share findings and identify research gaps in brain injury and long-term cognitive disorders in athletes.

Design: The conference concluded with a discussion of ways to best mitigate the risk of concussion and repetitive traumatic brain injury (RTBI).

Setting and participants: The summit was supported by an unrestricted educational grant from the National Football League and featured a diverse group of experts from multiple disciplines.

Interventions: N/A.

Main outcome measures: This discussion resulted in unanimous agreement supporting six consensus statements aimed at enhancing player safety and health.

Results: These consensus statements are as follows: (1) Eliminate intentional and avoidable head impacts in sports practices and games. (2) Encourage policies that limit the number, duration, and intensity of head impacts during sports practices and games. (3) Reinforce proper and safer techniques to avoid head contact at all levels of play. (4) Implement rules that reduce and penalize intentional and

avoidable contact to the head and neck. (5) Investigate specific clinical signs and symptoms associated with chronic traumatic encephalopathy neuropathology through further research. (6) Improve the criteria for traumatic encephalopathy syndrome through continued research.

Conclusions: These consensus statements highlight opportunities to advance the understanding and prevention of concussions and RTBI, by emphasizing the need for ongoing research and policy changes to safeguard athletes' health and well-being. Implementation of these changes would reduce the burden of brain injuries in sports, promoting a safer environment for athletes at all levels.

Questioning the rules of engagement: A critical analysis of the use of limb symmetry index for safe return to sport after anterior cruciate ligament reconstruction. British Journal of Sports Medicine, 59(6): 376-384, 2025.

Objective: To evaluate the association between limb symmetry index (LSI) in quadriceps and hamstrings strength together with hop tests, as a proxy of recovery, and the deviation from being symmetrical (LSI 100%), with a safe return to sport (RTS) after anterior cruciate ligament reconstruction (ACL-R).

Methods: Athletes between 15 and 30 years old with a preinjury Tegner activity level ≥6 were eligible for inclusion. Data were extracted from a rehabilitation-specific registry, Project ACL (Gothenburg, Sweden) at the time of or after RTS for each athlete. The outcome of interest in this study was a safe RTS-not suffering a second ACL injury within 2 years from RTS-addressed by using the LSI in five different ways with results from the test battery which include five tests of muscle function at or after the time of RTS. Logistic regression analyses were performed with safe RTS as a dependent variable and presented with ORs and 95% CIs.

Results: In total, 233 athletes (51.1% women) were included. The best-differentiating cut-offs for strength and hop tests showed non-significantly poor discriminatory ability between athletes who had a safe RTS and those who did not (Youden J 0.09-0.24 and area under the curve 0.50-0.59). Athletes who had ≥80% or ≥85% LSI had significantly lower odds of safe RTS compared with athletes who did not meet the cut-offs of safe RTS (OR=0.32 (95% CI 0.12 to 0.87) and OR=0.39 (95% CI 0.18 to 0.84), respectively). There was no effect of a 1% increase in LSI or deviation from 100% symmetry on safe RTS.

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Conclusion: The use of LSI from tests of muscle function to determine safe RTS after ACL-R, that is, RTS without sustaining a second ACL injury within 2 years, cannot differentiate between athletes who had a safe RTS and those who did not, regardless of whether LSI was used as cut-offs, incremental or as deviation from symmetry. Thus, it is of clinical importance that clinicians do not solely rely on the LSI to clear athletes for RTS.

Relationship among depression, fatigue, and sleep after traumatic brain injury: The role of physical exercise as a non-pharmacological therapy. Experimental Neurology, 386: 115156, 2025.

Abstract - Traumatic brain injury (TBI) is a burdensome condition frequently associated with an increased risk of psychiatric disorders. Although the exact molecular signaling pathways have not yet been fully defined, the compromised integrity of functional brain networks in regions such as the prefrontal cortex and anterior cingulate cortex has been linked to persistent symptoms, including depression, fatigue, and sleep disorders. Understanding how TBI affects neural physiology enables the development of effective interventions. One such strategy may be physical exercise, which promotes neural repair and behavioral rehabilitation after TBI. However, there are caveats to consider when interpreting the effects of physical exercise on TBI-induced mental health issues. This review will highlight the main findings from the literature investigating how different physical exercise protocols affect the progression of TBI-induced depression, fatigue, and sleep disturbances. Furthermore, we aim to explore potential neurobiological pathways that explain how physical exercise influences depression, fatigue, and sleep following TBI.

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Return to sport, reinjury rate, and tissue changes after muscle strain injury: A narrative review. Translational Sports Medicine, Article ID: 2336376, 2024.

Abstract - A major challenge in sports medicine is to facilitate the fastest possible recovery from injury without increasing the risk of subsequent reruptures, and thus effective rehabilitation programs should balance between these two factors. The present review focuses on examining the role of different resistance training interventions in rehabilitation of acute muscle strain in the time frame from injury until return to sport (RTS), the rate of reinjuries, and tissue changes after injury. Randomized, controlled trials dealing with a component of resistance training in their rehabilitation protocols, as well as observational studies on tissue morphology and tissue changes as a result to muscle strain injuries, were included. The mean time for RTS varied from 15 to 86 days between studies (n = 8), and the mean rate of reinjury spanned from 0 to 70%. Eccentric resistance training at long muscle length and rapid introduction to rehabilitation postinjury led to significant improvement regarding RTS, and core-stabilizing exercises as well as implementing an individualized algorithm for rehabilitation seem to reduce the risk of reinjury in studies with a high rerupture rate. Independent of the rehabilitation program, structural changes appear to persist for a long time, if not permanently, after a strain injury.

Return to play criteria based on infrared thermography during anterior cruciate ligament rehabilitation in football players. Biology of Sport, 42(2): 161-167, 2025.

Abstract - Reconstruction of the anterior cruciate ligament (ACL) is associated with degenerative changes during the return-to-play (RTP) process. Infrared thermography, due to its usefulness in monitoring the state of tissues, could help establish physiological criteria for monitoring the knee joint. The aim of the study was to describe normative skin temperature (Tsk) asymmetry of the knee region of interest (knee ROI) from football players during the ACL RTP process. Thirty professional and semi-professional football players, both male and female, from three different European leagues (age: 26.84±3.46 years; height: 1.83±0.22 m; weight: 75.38±5.92 kg) were evaluated using thermography during their RTP process. A linear mixed model was then developed in which Tsk asymmetry was the dependent variable. Sex and Day were included as predictor variables with fixed effects. Athlete was

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included as a random effect with (Model 1 and Model 2) and without (Model 3) interaction with Day. Model 2 was fitted with a random intercept and random independent slopes, and Model 3 was fitted with a random intercept and random dependent slopes. The results showed the estimates of the model in which both predictor variables were significant. The asymmetry decreased during the days after surgery (-0.01 per day after surgery). The results might help to establish an objective criterion based on infrared thermography for monitoring the rehabilitation stages in the RTP of football players during one year of evolution. The assessment of the thermal asymmetries of the knee ROI during the RTP established a Tsk progression criterion. Medical and technical staff could implement an infrared thermography tool for knee monitoring between RTP stages.

Tackle techniques and characteristics associated with a concussion in tackling players in the National Football League. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2025.

Background: Concussions remain a major concern in football. A qualitative video review can provide contextual evidence supporting changes in rules and coaching techniques aimed at reducing the risk of players sustaining a concussion.

Purpose: To identify tackling techniques and characteristics associated with concussions to the tackling player.

Study design: Case-control study; Level of evidence, 3.

Methods: A qualitative review of 51 concussions sustained by tacklers in National Football League (NFL) games between the 2015 and 2019 seasons as well as 96 control tackles was performed using video from multiple perspectives. For each concussive case, 1 to 3 nonconcussive control tackles were matched based on player trajectory and closing speed as determined from on-field player tracking data from Next Gen Stats. Cases and controls were coded for 20 different tackle techniques and characteristics. The data were analyzed using conditional logistic regression models, and the results were expressed as odds ratios (ORs) with 95% confidence intervals (CIs) as well as adjusted and unadjusted P values.

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Results: Despite attempting to match cases and controls based on closing velocity, closing velocity was significantly associated with the concussion risk in every logistic regression model. Predictors that had the greatest effect on the increased risk of concussions were "tackler leading with helmet" (OR, 5.18 [95% CI, 1.87-14.33]; false discovery rate [FDR]-adjusted P = .0032) and "tackler primary loading" being the helmet as opposed to the shoulder (OR, 4.61 [95% CI, 1.41-15.05]; FDR-adjusted P = .0000). Important factors associated with a reduced concussion risk were the tackler's "head placed on correct side" of the ball carrier (OR, 4.17 [95% CI, 1.58-11.01]; FDR-adjusted P = .0088) and "ball carrier primary loading," with the torso having the lowest risk relative to all other body segments.

Conclusion: The less a tackler involved his helmet in a tackle, the lower his risk of sustaining a concussion. This study supports teaching players to place their heads on the correct side of the ball carrier (neither into the ball carrier nor into the path of the ball carrier) when tackling to reduce their risk of a concussion, and players should target the torso of the ball carrier as their primary contact point.

The benefits of research-embedded training camps in sport sciences. International Journal of Sports Physiology and Performance, Ahead of Print: 1-5, 2025.

Background: Sport scientists aim to enhance athletic performance by developing, implementing, and evaluating evidence-based strategies. However, a common challenge to maximize sport science's impact is the limited involvement of "research end users" (ie, athletes, coaches, and practitioners) in contextualizing these strategies. To address this, research-embedded training camps (RETCs) offer a collaborative opportunity bringing together all stakeholders to tackle real-world challenges by blending scientific insight with practical application for better knowledge translation or production.

Purpose: This invited commentary explores the behind-the-scenes aspects of conducting successful RETCs. We discuss opportunities and challenges involved, drawing on our practical experience with environmental stress interventions. Evidence: Implementing an RETC requires a tailored intervention, shaped by the athlete group, research questions, and available resources—essentially answering the who, what, why, when, and where questions. Key principles for success include an athlete-centered focus, a diverse multidisciplinary team, scientific rigor, balancing control with flexibility, and adaptability.

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A co-constructed design ensures that the intervention remains scientifically relevant and practical and directly addresses real-world needs of athletes.

Conclusion: Through the lens of environmental stress interventions, RETCs should prioritize practicality, collaboration, and athlete-focused approaches. Conducting interventions with athletes rather than on them can significantly enhance performance and support evidence-based training strategies (knowledge translation) and/or provide new actionable insights (knowledge production).

The effect of a flywheel hip extension vs a traditional hip extension exercise on hamstring strength. International Journal of Strength and Conditioning, 4(1): 1-9, 2024.

Abstract - The main aim of this present study was to compare the effect of a flywheel hip extension exercise versus a traditional gravity-dependent exercise on hamstring strength. Twenty U-20 male soccer players volunteered to participate in the study. None had experience with flywheel inertia training, but all had a minimum of 12 months traditional resistance training experience. The participants were randomly assigned to two groups: the flywheel Romanian deadlift and traditional Romanian deadlift groups. Resistance training was performed twice a week for six weeks. Both groups performed four sets of six working repetitions, with the flywheel group performing two extra submaximal pre-repetitions at a low intensity to initiate the rotational force of the flywheel. A significant main effect of time was found for both eccentric strength ($p = 0.006$) and 3RM ($p < 0.001$) tests. There was a significant time-by-group interaction for 3RM ($p = 0.02$) but not for eccentric strength ($p = 0.18$). Post hoc analyses showed a significant increase in eccentric hamstring strength at the end of the intervention compared to baseline in the flywheel group (13% change, 37 N, $p = 0.03$, $g = 0.51$) but not in the traditional group (5% change,14 N, $p = 0.282$, $g = 0.18$). Both groups showed significant increases in 3RM Romanian deadlift (flywheel group: 18% change,19 kg, $p = < .001$, $g = 1.07$; traditional group: 28 % change, 26 kg, $p = < .001$, $g = 0.99$). This study highlights the beneficial use of flywheel training for optimizing hamstring strength adaptation. Although both groups showed similar improvements in the 3RM Romanian deadlift, the flywheel group showed superior eccentric strength improvements.

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The efficacy of nutritional strategies and ergogenic aids on acute responses and chronic adaptations to exertional-heat exposure: A narrative review. Nutrients, 16: 3792, 2024.

Abstract - Global warming is attributed to an increased frequency of high ambient temperatures and humidity, elevating the prevalence of high-temperature-related illness and death. Evidence over recent decades highlights that tailored nutritional strategies are essential to improve performance and optimize health during acute and chronic exertional-heat exposure. Therefore, the purpose of this review is to discuss the efficacy of various nutritional strategies and ergogenic aids on responses during and following acute and chronic exertional-heat exposure. An outline is provided surrounding the application of various nutritional practices (e.g., carbohydrate loading, fluid replacement strategies) and ergogenic aids (e.g., caffeine, creatine, nitrate, tyrosine) to improve physiological, cognitive, and recovery responses to acute exertional-heat exposure. Additionally, this review will evaluate if the magnitude and time course of chronic heat adaptations can be modified with tailored supplementation practices. This review highlights that there is robust evidence for the use of certain ergogenic aids and nutritional strategies to improve performance and health outcomes during exertional-heat exposure. However, equivocal findings across studies appear dependent on factors such as exercise testing modality, duration, and intensity; outcome measures in relation to the ergogenic aid's proposed mechanism of action; and sex-specific responses. Collectively, this review provides evidence-based recommendations and highlights areas for future research that have the potential to assist with prescribing specific nutritional strategies and ergogenic aids in populations frequently exercising in the heat. Future research is required to establish dose-, sex-, and exercise-modality-specific responses to various nutritional practices and ergogenic aid use for acute and chronic exertional-heat exposure.

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Tired of ACL Injuries: A review of methods and outcomes of neuromuscular fatigue as a risk factor for ACL injuries. Biomechanics, 5: 11, 2025.

Background/Objectives: One potential risk factor that remains especially contentious in the anterior cruciate ligament (ACL) injury literature is the role of neuromuscular fatigue in ACL injury risk. Therefore, the purposes of this review are (i) to present the research and practical concepts of lower extremity neuromuscular fatigue; and (ii) to review the literature relating to neuromuscular fatigue as an ACL injury risk factor and mechanism.

Methods: A structured review was performed in the Medline database using a search strategy that included terms such as “anterior cruciate ligament injury” and “knee injuries” combined with terms such as “injury” and “fatigue”. Articles were included if they included young healthy participants (18–35) and made a comparison between non-fatigued and fatigued states that were assessed with at least one lower extremity biomechanical variable associated with ACL injury risk.

Results: Overall, there were 67 studies included, accounting for 1440 participants (627 male and 813 female) across a variety of sports and activities. Of these, 53 (79%) reported a post-fatigue change in the kinematics, kinetics, neuromuscular, and/or other (e.g., proprioceptive) outcomes that indicate that the participants would be at an increased risk of an ACL injury. The most common argument against fatigue as a risk factor is that ACL injuries do not tend to occur later in a game or season, when it is assumed that athletes would be most fatigued.

Conclusions: The evidence presented in this review suggests that localized neuromuscular fatigue is a risk factor, among multiple factors, for ACL injuries, providing another modifiable risk factor that should be considered when developing ACL injury risk reduction interventions.

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Tirzepatide in Sport: A comprehensive review of its metabolic impacts and potential applications for athletes. Quality in Sport, 37: 1-15, 2025.

Introduction: Tirzepatide, a dual agonist of glucose-dependent insulinotropic peptide (GIP) and glucagon-like peptide-1 (GLP-1) receptors, has emerged as a transformative metabolic therapy with exceptional efficacy in weight management and glycemic control. Its relevance in sports science is growing, with potential benefits in weight optimization, energy balance, and recovery. By influencing glucose homeostasis, fat oxidation, and systemic inflammation, tirzepatide may enhance athletic performance. This review examines its pharmacological effects, impact on body composition, and implications for endurance, strength, and recovery in athletes, while addressing ethical and regulatory considerations, including anti-doping concerns.

Purpose of the Work: To analyze tirzepatide's pharmacological actions, clinical effects, and potential applications in optimizing sports performance, recovery, and body composition.

Materials and Methods: A comprehensive review of scientific literature was conducted across databases including PubMed, Scopus, Web of Science, Embase, and Google Scholar. Search terms encompassed tirzepatide, GLP-1/GIP dual agonist, metabolic therapy, sports performance, body composition, endurance, and athlete recovery. Relevant studies were synthesized to compile findings.

Results: Tirzepatide provides significant metabolic benefits, such as enhanced fat oxidation, improved glucose regulation, and reduced systemic inflammation. Its effects on body composition include substantial fat mass reduction, particularly visceral adiposity, while preserving lean muscle mass. These attributes suggest its potential to enhance endurance via glycogen-sparing mechanisms and improve recovery after exercise. However, concerns about misuse in competitive sports underline the need for regulatory oversight. Further research is necessary to evaluate tirzepatide's long-term effects on athletic performance and address associated ethical challenges.

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Toxic leadership in high-performance sports and its consequences for mental health and performance: A scoping review. International Review of Sport and Exercise Psychology, Ahead of Print: 1-28, 2025.

Abstract - This scoping review aimed to identify the current state of knowledge on toxic leadership in high-performance sports and its consequences for mental health and performance at individual, interpersonal, and organizational levels. Abstract screening (n = 768) and full-text review (n = 83) resulted in the inclusion of 26 studies, predominantly conducted in North America and Europe. Studies used inconsistent definitions, theoretical frameworks, and methodological approaches to study darker or toxic leadership. Several antecedents and maintaining factors to toxic leadership were identified, such as individual leader characteristics, a performance-oriented culture, status/power structures, and a lack of reporting processes or consequences within organizations. The empirical literature provided initial support for the negative consequences of toxic leadership on athletes' mental health, injuries, and motivation to remain in high-performance sports. Few studies have investigated the consequences of toxic leadership on sports organizations, staff or coaches. We conclude that research on toxic leadership in high-performance sports is in its early stages, with an underdeveloped understanding of how problematic leadership can be detected, prevented, and addressed to minimize its impact in sports. Researchers, governing bodies, and sports organizations must intensify their efforts to comprehend the emergence and persistence of toxic leadership within high-performance sports environments.

Weighted approaches to assessing injury burden in team sports. Journal of Sports Analytics, 10(4): 1-9, 2025.

Abstract - Injuries have a large impact on a team's success during the season. Typically, teams with more injuries, especially those affecting the most valuable players, will perform worse than similar teams with fewer injuries. A common way to quantify a team's injury burden is by simply counting the number of injuries or games missed due to injury. However, these measures do not account for the individual value of the players injured. We assess two frameworks for how injuries may be weighted by player value. We propose a new position-weighted metric based on franchise tag values to measure injury burden in American football, called weighted Adjusted Games Lost (wAGL), and we test another player-level

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metric, Wins Above Replacement-Adjusted Injuries Lost (WAIL). WAIL uses a proprietary calculation of player Wins Above Replacement from Pro Football Focus. By comparing these measures to a gold standard of injury burden using pre- versus end-season betting lines and team strength estimates, we find that wAGL is a better measure of injury burden than current alternatives across all team-seasons considered.